

TASK 2: Sentiment Analysis using Machine Learning and LSTM

Name: Sivaranjani.S

Registration Number: INBT019473

Internship Program: iNeuBytes(AI)

Introduction

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine whether a piece of text expresses a positive or negative opinion. It is widely used in product reviews, social media analysis, customer feedback systems, and opinion mining.

The objective of this project is to classify IMDB movie reviews into Positive and Negative sentiments using both Machine Learning and Deep Learning techniques.

Objectives

- Build sentiment classification models.
- Compare Logistic Regression, SVM, and LSTM.
- Evaluate model performance.
- Identify the best-performing model.

Dataset

The IMDB Movie Reviews dataset contains movie reviews labeled as Positive or Negative.

Preprocessing

The dataset was cleaned using lowercase conversion, punctuation removal, tokenization, stopword removal, TF-IDF vectorization for ML models, and tokenization with padding for the LSTM model.

Machine Learning Models

Logistic Regression

Logistic Regression was used as a baseline classifier for binary sentiment classification.

Support Vector Machine (SVM)

SVM was trained using TF-IDF features and achieved better performance than Logistic Regression by identifying an optimal decision boundary.

Deep Learning Model

LSTM

The LSTM network was trained on padded sequences generated from the tokenized reviews. The model automatically learned contextual information from text and produced the highest classification performance.

Evaluation Metrics

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Results

The LSTM model achieved the best overall performance, followed by SVM and Logistic Regression. Its ability to capture sequential relationships within text contributed to higher accuracy and better generalization.

Conclusion

The project successfully implemented both Machine Learning and Deep Learning approaches for sentiment analysis. Comparative evaluation demonstrated that the LSTM model outperformed the traditional Machine Learning models, making it the most suitable approach for sentiment classification on the IMDB dataset.